

A3-PS033-TC378

DNS-Based Discovery of Diagnostic Services

Chris Itty Biju and Alan Jayan Joseph

Faculty mentor: Revathi BS

Karunya University of Technology and Sciences

Problem Description

Changing service locations

Hardcoded IP addresses and ports become outdated when diagnostic endpoints move.

Unverified connection details

The client needs to detect altered DNS data and reject metadata it cannot interpret.

Proposed approach

Discover service details through signed DNS records, then authenticate before invocation.

Background

DNS and SVCB

Find the service hostname, port and capability metadata.

DNSSEC

Check the origin and integrity of the published DNS records.

TLS and service credentials

Verify the endpoint identity and the caller's permission.

Valid DNS metadata alone grants no service access.

Literature & RFC Review

DNS-AID draft -02

Pinned proposal for known-service discovery and security boundaries. Work in progress.

RFC 9460

SVCB ServiceMode records carry connection parameters and mandatory metadata keys.

RFC 4033, RFC 4034 and RFC 4035

DNSSEC requirements, signed record formats and validation procedures.

Proposed Solution

One signed test zone

A publisher advertises service connection details in diag.test using SVCB and address records.

Three mock diagnostic services

Latency, traceroute and health services return clearly labelled sample results.

One discovery and invocation client

The client validates DNSSEC, metadata and TLS before sending a service-specific token.

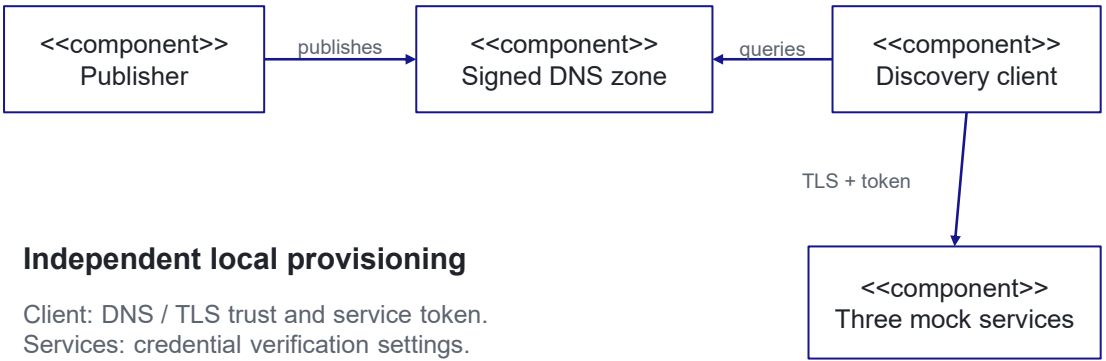
Local prototype: real security checks, simulated diagnostic measurements.

Optimization Proposed

Area	Manually configured setup	Proposed discovery workflow
Endpoint change	Update each stored endpoint	Publish new signed connection details
Metadata integrity	No signature check by default	Reject invalid or unsigned DNS answers
Access control	Credentials configured separately	Keep credentials separate from discovery
Repeated lookups	No discovery cache involved	Evaluate TTL-aware caching later

Port migration already works in the lab. Caching and performance gains remain to be evaluated.

Solution Architecture & Design

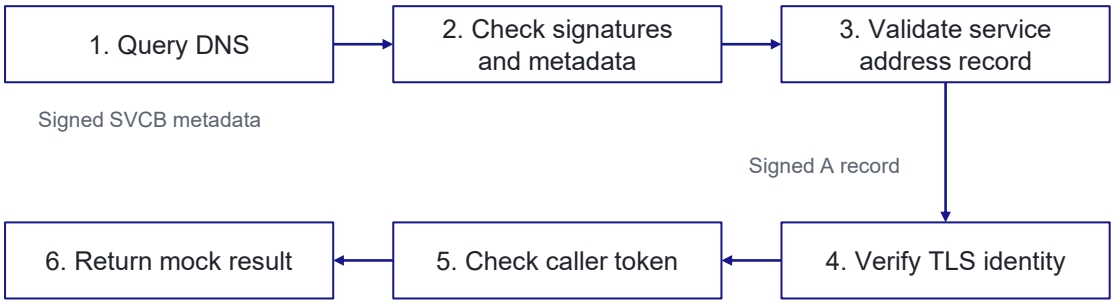


Independent local provisioning

Client: DNS / TLS trust and service token.
Services: credential verification settings.

DNS integrity, endpoint identity and service access remain separate checks.

Data Flow Diagram



Invalid signatures or metadata stop discovery.
Connection failures or rejected credentials stop invocation.

Deployment Timeline

8-10 October: preparation

Review scope with the mentor, finalize the proposal and resolve the GitHub setup questions.

11-13 October: Level 1 materials

Record each member, upload the PDF and video link, and confirm repository access and transfer.

14 October: Level 1 deadline

Complete steps 1-7. During 16-31 October, continue mentoring, code and demo delivery.

13 October is our suggested readiness target, not an organizer-issued deadline.

Bibliography

[1] DNS-AID draft -02

Mozley et al., 27 May 2026. Individual Internet-Draft.

[2] RFC 9460

Service Binding and Parameter Specification via the DNS.

[3] RFC 4033, RFC 4034 and RFC 4035

DNSSEC requirements, resource records and protocol changes.

[4] AIORI-3 organizer materials

PS-033, presentation structure and online hackathon guidelines.

Draft ID: draft-mozleywilliams-dnsop-dnsaid-02

Thank You

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**One signed zone. Three mock services.
A client that checks before it invokes.**

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